

ELECTRIC CHARGE & COULOMB'S LAW

Q.1 Define and explain electric ' charge, hence differentiate between conductors and insulators?

ELECTRIC CHARGE

The strength of electrical interaction of a particle with objects around it depends on its electric charge. Charges with same sign repel each other and charges with opposite signs attract each other. The object with equal amounts of the two kinds of charge is called electrically neutral. The object with imbalance the two kinds of charge is called electrically charged.

All materials including solids, liquids and gases contain two basic particles of electric charge called electron and proton. The electron has negative polarity whereas proton has positive polarity.

It is the arrangement of electrons and protons that determines the electrical properties of all the substances. The sharing of electrons takes place when one material is rubbed against the other. The material that loses the electrons is called positively charged. The material that gains the electrons is called negatively charged.

EXAMPLE

The rod gets positively charged when it is rubbed against the silk whereas silk is charged negatively.

UNIT OF CHARGE

The SI unit of charge is called coulomb. The charge of 6.25×10^{18} electrons or protons is equal to one coulomb charge. The submultiple units of charge are μC , nC and pC etc.

The study of electric charges at rest under the action of electric forces is called electrostatics.

TYPES OF CHARGES

The charge is of two types: positive charge and negative charge. The similar charges repel each other with electric force. The unlike charges attract each other with electric force.

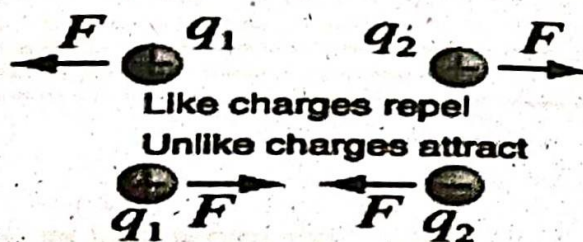
There are

two types of charges called positive and negative.

Like charges repel each other.

Unlike charges attract each other.

There are



POLARITY OF CHARGE

The charges of same polarity repel each other while charges of opposite polarity attract each other. The positive and negative labels to charged materials are arbitrary and assigned by the Franklin.

CHARGE ON ELECTRON

The magnitude of charge on electron is 1.6×10^{-19} C. It is measured by Milliken.

The mass of an electron is 9.1×10^{-31} kg.

CHARGE ON PROTON

The magnitude of charge on proton is 1.6×10^{-19} C.

The mass of proton is 1.67×10^{-27} kg.

CHARGE ON NEUTRON

The neutron has no charge. Its mass is approximately equal to mass of proton.

POLES OF CHARGE

The charge forms poles.

1. An isolated single charge is called an electric monopole.

2. Equal positive and negative charges having constant separation between them is called electric dipole.

3. Equal positive and negative charges having constant separation between them is called electric dipole.

4. Two oppositely directed dipoles close to each other are called an electric quadrupole.

You can continue this process to any number of poles, but dipoles and quadrupoles are important because they find significant application in physical phenomenon.

CONDUCTORS

The materials through which charges can flow easily are called conductors. Some properties of conductors are:

[CHAPTER ELECTRIC CHARGE & COULOMB'S LAW

1. O The copper and aluminum are good of conductors.
2. T O The free electrons are mobile and positive ionic cores are immobile within conductor.

11

3

O Each atom of a typical conductor may contribute one conduction electron. It means a conductor on average has 10^{23} conduction electrons/cm.

O The conduction band and valence band overlaps, in conductors so electrons can easily move.

O The resistivity of conductors is of the order of $10^{-8} \Omega\text{-m}$. The resistivity of Cu is $1.77 \times 10^{-8} \Omega\text{-m}$.

SEMICONDUCTORS

The partial conductors are called semiconductors.

The partial conductors are called semiconductors.

1. The well-known examples of semiconductors are silicon and germanium. A typical
2. semiconductor has 10^{12} conduction electrons/cm.
3. D The resistivity of semiconductor germanium is $0.5 \Omega\text{-m}$.

INSU

INSULATORS

insulators.

1. The materials through which charges cannot flow are called
 2. O The electrons are tightly bound in such materials.
 3. O The well-known insulators are glass and plastic.
 4. O The insulator has only one conduction electron/cm.
 5. O The resistivity Of insulators is of the Order of $10^{16} \Omega\text{-m}$. The resistivity of glass is $2 \times 10^{11} \Omega\text{-m}$.
- O The energy gap between conduction band and valence band in insulator is very, high.
The electrons cannot jump from valence band to conduction.

Q.2 Discuss quantization of electric charge in detail?

QUANTIZATION OF CHARGE

The electric charge was thought to be a continuous fluid in Benjamin Franklin's day. Nowadays we know that fluids such as air and water are not continuous but are made of atoms and molecules. Experiment shows that electrical fluid is also not continuous but is made up of multiples of a certain elementary charge.

Any positive or negative charge q can be written as

$$q = ne$$

Where $n = \pm 1, \pm 2, \pm 3$ - -and $e = 1.6 \times 10^{-19} \text{ C}$

The elementary charge e is one of the important constants of nature. The electron and proton both have a charge of magnitude e . The quarks which are constituent particles of protons and neutrons have charges of $\pm \frac{1}{3}e$ or $\pm \frac{2}{3}e$ but they cannot be detected

- individually. That is why we do not take their charges to be the elementary charge.

You often see phrases such as "the charge on a sphere;" "the amount of charge transferred," and "the charge carried by the electron". These phrases suggest that

charge is a substance. You should however keep in mind what is indeed. Particles are the substance and charge happens to be one of their properties just as mass is.

4

We say a physical quantity such as charge is quantized when it can have only discrete values rather than any value. For example, it is possible to find a particle that has no charge at all or a charge of $+1e$ or $-6e$ but not a particle with a charge of say $3.57e$.

The quantum of charge is small. For example, in an ordinary 100 W light bulb about 10^{19} elementary charges enter the bulb every second and just as many leave. However, the graininess of electricity does not show up in such large-scale phenomena (the bulb does not flicker with each electron) just as you cannot feel the individual molecules of water with your hand.

Q.3 Discuss conservation of electric charge in detail with examples?

CHARGE CONSERVATION

The positive charge appears on the rod when you rub a glass rod with silk. The measurement shows that negative charge of equal magnitude appears on the silk. This suggests that rubbing does not create charge but only transfers it from one body to another body. This hypothesis of conservation of charge was first put forward by Benjamin Franklin. Some examples of charge conservation are

RADIOACTIVE DECAY

The conservation of charge occurs in radioactive decay of nuclei in which a nucleus transforms into a different type of nucleus. For example, a uranium-238 nucleus transforms into a thorium-234 nucleus by emitting an alpha particle.



The parent nucleus ${}_{92}\text{U}^{238}$ contains 92 protons having charge of $+92e$. The daughter nucleus ${}_{90}\text{Th}^{234}$ contains 90 protons having charge of $+90e$. The emitted alpha particle ${}_2\text{He}^4$ contains 2 protons having charge of $+2e$. The total charge is $+92e$ before and after the decay. Thus charge is conserved.

The total number of protons and neutrons before decay are 238 and after decay are $234 + 4 = 238$, Hence total number of protons and neutrons are also conserved in decay process.

[CHAPTER ELECTRIC CHARGE & COULOMB'S LAW ANNIHILATION OF MATTER

The charge conservation occurs in annihilation process when an electron having charge $-e$ and its antiparticle called positron having charge $+e$ annihilates in such a way that they transform into two gamma rays.

The net charge of the system is zero both before and after the event. Hence charge is conserved.

PAIR PRODUCTION

The charge is also conserved in pair production that converse process of annihilation. In this process a gamma ray transforms into an electron and a positron

The net charge of the system is zero both before and after the event. Hence charge is conserved.

Q.4. State and explain Coulomb's law in detail?

COULOMB'S LAW

The electric force between two charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

EXPLANATION

Consider two point charges q_1 and q_2 separated by distance r . The magnitude of electric force F_e according to Coulomb's law is

On combining together

$$\frac{1}{r^2}$$

Where k is constant of proportionality and depends upon the nature of medium between charges and system of units used to measure force, charge and distance. If medium between charges is free space, value of k is written as

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N-m}^2/\text{C}^2$$

Where ϵ_0 is an electrical constant called permittivity of free space having value $8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2$.

The Coulomb's force in free space is given as

$$\frac{1}{r^2}$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

The Coulomb's force is a mutual force. It means force F_{m1} exerted by charge q_1 on charge q_2 is equal in magnitude but opposite in direction to force F_{12} exerted by charge q_2 on charge q_1

electric force between two charges in vector form is written as

$$q_2$$



$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{21}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

EFFECT OF MEDIUM

The presence of a medium called dielectric between charges reduces the electric force between them as compared to the force in free space by certain factor that is constant for the given dielectric.

This constant is called relative permittivity and denoted by ϵ_r . The values of relative permittivity of different dielectrics are given in table.

Material	Vacuum	Air(1atm)	Ammonia (liquid)	Bakelite	Benzene	Germanium	Glass
ϵ_r	1	1.0006	22-25	5-16	2.284	16	4.8-10
Material	Mica	Paraffined paper	Plexiglas	Rubber	Teflon	Transformer oil	Water (distilled)
ϵ_r	3-7.5	2	3.40	2.94	2.1	2.1	78.5

The Coulomb's force in medium of relative permittivity ϵ_r is given as

$$F_{\text{med}} = \frac{F_c}{\epsilon_r} = \frac{1}{4\pi\epsilon_0 \epsilon_r} \frac{q_1 q_2}{r^2}$$

USES OF VECTOR FORM OF COULOMB'S LAW

The vector form has following uses:

- ① It tells whether force is attractive or repulsive
- ② The resultant force can be calculated from vector sum of forces when assembly of charges is under consideration.

ELECTRIC FORCE OF N-CHARGES

Consider assembly of n-charges. The total force exerted on charge "q" placed among n- point charges can be calculated by superposition principle as

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n$$

$$\vec{F} = kq \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$

Where $i = 1, 2, 3, 4, \dots, n$

SIGNIFICANCE OF COULOMB'S LAW

The Coulomb's law describes completely how electrons are attracted by nucleus in atom, how atoms form molecules and how atoms and molecules develop into solids or liquids.

permittivity of different dielectrics are given in table.

The Coulomb's force in medium ϵ_r is given as



liquids.

The Coulomb's law not only holds for macroscopic charged particles, it is equally applicable to microscopic particles like electron, proton. However, law is not applicable when charged particles move close to velocity of light and inside the nucleus under certain conditions because there exist nuclear force rather than Coulomb's force.

Q.5. Explain how Coulomb's law is verified experimentally ?

EXPERIMENTAL VERIFICATION

[CHAPTER ELECTRIC CHARGE & COULOMB'S LAW

The charges of same polarity attract each other with a force called APPARATUS

Coulomb measured charges using apparatus called torsion balance consists of two metallic spheres a and a at its diagram.

The rod is suspended from the middle with a fiber. A small mirror M is attached to the fiber and beam of light reflected from mirror falls on the scale.

WORKING

An insulated rod P carrying a small charged sphere b equal in size to sphere a is brought near sphere a. Both spheres share charge $q_1 = q_2 = q$ when sphere b is touched with sphere a. The charge on sphere b can be further divided into $q/2$; $q/4$ by touching it with the spheres of equal sizes using same method.

By keeping the distance between sphere a and sphere b fixed, we find that twisting (electric force) in fiber is directly proportional to the magnitude of the charges on sphere a and sphere b.

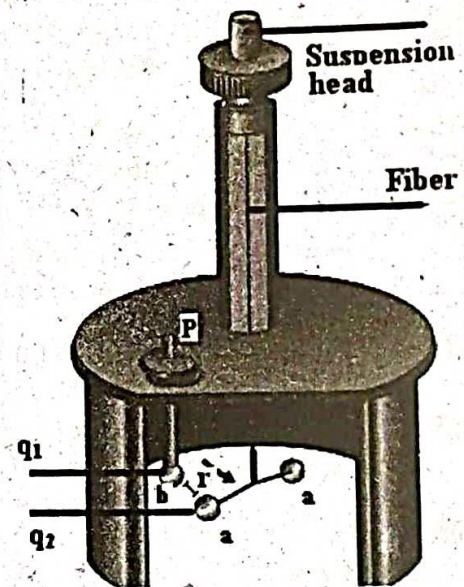
The twisting (electric force) is found inversely proportional to square of the distance between the charged spheres by changing the distance between them.

olarity repel each other and charges of opposite polarity called electrostatics force.

electric force between d torsion balance. The horizontal rod carrying at its ends as shown in

rom the middle with a attached to the fiber and mirror falls on the scale.

rying a small charged here a is brought near charge $q_1 = q_2 = q$ h sphere a. The charge divided into $q/2$, $q/4$ by equal sizes using same



SHORT QUESTIONS

Q:1 The quantum of charge is $1.6 \times 10^{-19} \text{ C}$, Is there a corresponding quantum of mass?

Ans: There is no quantum of mass.

Q:2 What does it mean to say that a physical

quantity is (a)

quantized or (b) conserved. Give Ans. CHARGE

When a physical

some examples.

QUANTIZATION

quantity exists in discrete packets, it is called quantized.

EXAMPLES

Charge, energy of photons, frequency, of vibrations Of stretched string and angular momentum of electron in orbits are some examples of quantized physical quantities.

CHARGE CONSERVATION

A physical quantity is called conserved when total amount of physical quantity remains constant before change and after change

EXAMPLES

Charge and energy are conserved 'in pair production and annihilation of matter.

Q:3 An electron charge $-e$ circulates around a helium nucleus of charge $+2e$ in helium atom. Which particle exerts the larger force on other?

Ans. The electric force between two charged bodies is action reaction force which is equal in magnitude but opposite in direction. The force exerted by electron on nucleus is equal to force exerted by nucleus on electron.

Q:4 Would it have any important difference if Benjamin Franklin had chosen, in effect, to call electrons positive and protons negative.

Ans. It makes no any important difference if Benjamin Franklin had chosen, in effect, to call electrons positive and protons negative.

Q:5 A solution of copper sulphate is a conductor. What particles serve as the charge carriers in this case.

Ans. The particles Cu^{+} and SO_4^{-} serve as charge carrier in solution of copper sulphate.

Q:6 Compare gravitational force and Coulomb force between two electrons. Which is greater?

Ans. The gravitational force between two electrons is G

$$\frac{m_e m_e}{r^2}$$

The Coulomb force between two electrons is

$$\frac{e^2}{4\pi\epsilon_0 r^2}$$

The ratio is

$$\frac{G m_e^2}{\frac{e^2}{4\pi\epsilon_0 r^2}} = \frac{4\pi\epsilon_0 G m_e^2}{e^2}$$

$$= 9 \times 10^{-42} \times (1.6 \times 10^{-19})^2$$

$$6.67 \times 10^{-11} (9.1 \times$$

It shows that $F_C > F_g$,

1)

9

SAMPLE PROBLEMS

p. 1 A penny being electrical neutral contains equal amount of +ve and —ve charge. What is magnitude of charge?

Sol.

DATA

Atomic no. of copper $z = 29$

Charge on copper penny — $q = NZe$

No. of atoms in penny

Charge of each atom —

Standard mass of penny = 3.11 g

Molecular mass of penny = 63.5 g/mole

CALCULATIONS

One mole of Cu has atoms = n

moles of Cu have atom's

Also n -moles of Cu atoms — Total no. of atoms =

$$\frac{3.11 \times 6.02 \times 10^{23}}{63.5}$$

$$2.95 \times 10^{22} \text{ atoms}$$

$$\text{Charge on penny} = NZe = 2.95 \times 10^{22} \times 29 \times 1.6 \times 10^{-19} \text{ C}$$

$$1.37 \times 10^5 \text{ C}$$

$$\text{Magnitude of +ve charge} = \text{Magnitude of —ve charge} = 1.37 \times 10^5 \text{ C}$$

P.2 The copper penny contains both +ve charge and —ve charge and each of magnitude $1.37 \times 10^5 \text{ C}$. Suppose that these charges could be concentrated into two bundles, held 100m apart. Find attractive force?

sol.

DATA

$$1.37 \times 10^5 \text{ C}$$

$$q_2 = 1.37 \times 10^5 \text{ C}$$

$$100 \text{ m}$$

F

CALCULATIONS

10

$$9 \times 10^9 \times (100)$$

$$10^5$$

$$1016 \text{ N}$$

P.3 The average distance between an electron and a proton in hydrogen atom is $5.3 \times 10^{-11} \text{ m}$. (a) Find the electrostatic force? (b) Find gravitational force

Sol.

DATA

$$1.6 \times 10^{-19} \text{ C}$$

$$q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$r = 5.3 \times 10^{-11} \text{ m}$$

$$m_1 = 9.1 \times 10^{-31} \text{ kg}$$

$$m_2 = 1.67 \times 10^{-27} \text{ kg}$$

CALCULATIONS

$$\frac{(1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = 8.2 \times 10^{-8} \text{ N}$$

(b)

$$\frac{6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.67 \times 10^{-27}}{(5.3 \times 10^{-11})^2} = 3.6 \times 10^{-47} \text{ N}$$

P.4 The nucleus of iron atoms has radius $4 \times 10^{-15} \text{ m}$ and have 26 protons. What repulsive force acts between two protons if they are separated by one radius? sol.

DATA

$$1.6 \times 10^{-19} \text{ C}$$

$$r = 4 \times 10^{-15} \text{ m}$$

$$1.6 \times 10^{-19} \text{ C}$$

CALCULATIONS

[CHAPTER ELECTRIC CHARGE & COULOMB'S LAW

$$\frac{10^9 \times (1.6)^2}{(4 \times 10^{15})^2} = 14.4 \text{ N}$$



EXERCISE PROBLEMS

P.1. A point charge of $+3.12 \times 10^{-6} \text{ C}$ is 12.3 cm from second charge $1.48 \times 10^{-6} \text{ C}$. Calculate magnitude of force?

Sol.

DATA

$$q_1 = 3.12 \times 10^{-6} \text{ C}$$

$$q_2 = 1.48 \times 10^{-6} \text{ C}$$

$$r = 12.3 \times 10^{-2} \text{ m}$$

$$F = ?$$

CALCULATIONS

$$F = \frac{kq_1q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times 3.12 \times 10^{-6} \times 1.48 \times 10^{-6}}{(12.3 \times 10^{-2})^2} = 2.748 \text{ N}$$

P.2. What must be distance between point $q_1 = 26.3 \mu\text{C}$ and charge $q_2 = 47.1 \mu\text{C}$ in order that force between them is 5.66 N?

Sol.

DATA

$$q_1 = 26.3 \times 10^{-6} \text{ C}$$

$$q_2 = 47.1 \times 10^{-6} \text{ C}$$

$$F = 5.66 \text{ N}$$

$$r = ?$$

CALCULATIONS

$$F = \frac{kq_1q_2}{r^2}$$

$$r = \left(\frac{kq_1q_2}{F} \right)^{1/2}$$

$$r = \left(\frac{9 \times 10^9 \times 26.3 \times 10^{-6} \times 47.1 \times 10^{-6}}{5.66} \right)^{1/2} = 1.4 \text{ m}$$

P.3. In the return stroke of typical lighting bolt, a current 2.5×10^4 A flows for 20 μ s. How much charge is transferred?

sol.

DATA

$$I = 2.5 \times 10^4 \text{ A} \quad t = 20 \times 10^{-6} \text{ s}$$

q

CALCULATIONS

$$q = I \times t = 2.5 \times 10^4 \times 20 \times 10^{-6} = 0.5 \text{ C}$$

P.4. Two equally charged particles held 3.20 mm are released from rest, the initial acceleration of first particles is 7.22 m/s^2 and that of second particle is 9.16 m/s^2 . The mass of first particle is $6.31 \times 10^{-7} \text{ kg}$. (a) Find mass of second particle (b) The magnitude of common charge?

sol.

DATA

For first particle:

$$a_1 = 7.22 \text{ m/s}^2$$

$$m_1 = 6.31 \times 10^{-7} \text{ kg}$$

CALCULATIONS

$$F = m_1 a_1$$

$$F = 6.31 \times 10^{-7} \times 7.22 = 4.55 \times 10^{-6} \text{ N}$$

(a)

F

$$4.55 \times 10^{-6} = m_2 \times 9.16$$

$$m_2 = 4.96 \times 10^{-7} \text{ kg}$$

(b)

First particle exerts the same force on second charged particle q

$$3.2 \times 10^{-3} \text{ m}$$

$$F = 4.55 \times 10^{-6} \text{ N}$$

$$q$$

11

13

CALCULATIONS

$$F$$

$$q$$

$$4.55 \times 10^{-6} \times (3.2 \times 10^{-3})^2$$

$$7.1 \times 10^{-11} \text{ C}$$

P.5. Two identical conducting spheres, 1 and 2, carry equal amounts of charge and are fixed a distance apart large compared with their diameters. They repel each other with electric force 88mN. Suppose now that a third identical sphere 3 having insulating handle and initially uncharged, is touched first to sphere 1, then to sphere 2, and finally removed. Find the force between sphere 1 and 2 now. Sol.

DATA

$$- 88 \times 10^{-3} \text{ N}$$

Let the charge on sphere 1 and 2 is q and separated by distance r

$$F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = 10^{-3} \text{ N}$$

When uncharged identical sphere 3, having insulating handle is touched to sphere 1 having charge q . The charge of each sphere becomes

$$\frac{q}{2}$$

Now sphere 3 of charge $\frac{q}{2}$ is touched to sphere 2 having charge q .

The charge of sphere 2 and sphere 3 becomes $\frac{3q}{4}$

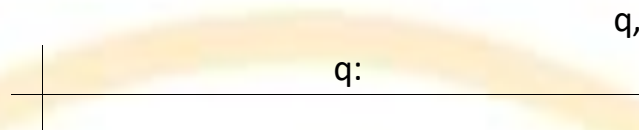
•The force between sphere 1 and sphere 2 is

P.6. Three charged particles lie on a straight line and separated by distance d such

14

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

P.6. Three charged particles lie on a straight line and separated by distance d such that charges q_1 and q_2 are fixed. Charge q_3 which is free to move and is found in equilibrium under the action of electric forces. Find q_1 in terms of q_2 .



Sol.

The charge is in equilibrium if magnitude of F_{13} is equal to F_{23} but opposite in direction

$$F_{13} = -F_{23}$$

$$q_1 = -4q_2$$

P.7. Each of two small spheres is charged positively, repelled by 1.19 N force. The total charge being 52.6×10^{-6} C on spheres are 1.94 m apart calculate charge on each sphere?

Sol.

DATA

$$52.6 \times 10^{-6} \text{ C}$$

$$52.6 \times 10^{-6} \text{ C}$$

$$1.19 \text{ N}$$

$$1.94 \text{ m}$$

$$1.19 \times (1.94)^2 q^2$$

Use quadratic formula

CALCULATIONS

$$10^9 \times 52.6 \times 10^{-6} \left| \begin{array}{c} 2 \\ 2 \end{array} \right|$$

-

$$(1.94)$$

$$9 \times 10^9 \times (52.6 \times 10^{-6} q^2 - q^2)$$

$$52.6 \times 10^{-6} + 2.79 \times 10^{-5}$$

$$2$$

$$52.6 \times 10^{-6} + 2.79 \times 10^{-5}$$

2

IJ

15

P.8. Two fixed charges $1.07 \times 10^{-6} \text{ C}$ and $-3.29 \times 10^{-6} \text{ C}$ are 61.8 cm apart where may a third charge be located so that no net force act on it? sol.

DATA

$$q_1 = 1.07 \times 10^{-6} \text{ C} \quad q_2 = -3.29 \times 10^{-6} \text{ C} \quad d = 61.8 \times 10^{-2} \text{ m} = 61.8 \text{ cm}$$

CALCULATIONS

Let at distance x from $+q_1$ charge a third charge q is placed. The electric force between q and q_1 is

The electric force between q and q_2 is

$$F_{12} = k \frac{q_1 q_2}{d^2}$$

Net force on charge q will be zero if direction of forces opposite but equal in magnitude

$$F_{1q} = F_{2q}$$

$$22.47 \text{ cm}$$

P.9. Find total charge in coulombs of 75 kg of electrons?

sol.

DATA

Mass of electrons 75 kg

Charge on electrons — CALCULATIONS

$$9.1 \times 10^{-31} \text{ kg electron has charge } e = 1.6 \times 10^{-19} \text{ C}$$

$$1.6 \times 10^{-19}$$

9.1×10

75 kg electron has charge $73 \times 1.75 \times 10^{11}$ $1.3 \times 10^{13} \text{ C}$

16

P.IO. In a crystal of salt an atom of sodium transfers one of its electrons to atom of chlorine. The resulting Na⁺ and Cl⁻ ion attracts each other. Calculate force of attraction if $r = 282 \text{ pm}$?

Sol.

DATA

+1. $1.6 \times 10^{-19} \text{ C}$

Charge on Na

Charge on Cl⁻ q_2 $1.6 \times 10^{-19} \text{ C}$

282 pm

CALCULATIONS

$$F = \frac{10(2.82 \times 10^{-9} \times (1.6 \times 10^{-19})^2)}{(282 \times 10^{-12})^2} = 2.89 \times 10^{-5} \text{ N}$$

P.II.' Electrostatic force between two identical ions that are separated by $5 \times 10^{-10} \text{ m}$ is $3.7 \times 10^{-9} \text{ N}$. (a) Find charge on each ion. (b) How many electrons are missing from each ion?

Sol.

DATA

q

$5 \times 10^{-10} \text{ m}$

$3.7 \times 10^{-9} \text{ N}$

CALCULATIONS

1

1.6×10

charge has
electrons = 1
electron

1

1C charge
has
electrons —

19

$$\frac{3.7 \times 10^{-9} \times (5 \times 10^{-10})^2}{109}$$

109

1.6×10^{19}

$$3.2 \times 10^{19} \text{ C} \quad \begin{array}{l} 3.2 \times 10^{19} \text{ charge has electrons} \\ 1.6 \times 10^{-19} \end{array} \quad \begin{array}{l} 2 \text{ electrons} \\ -19 \end{array}$$

[CHAPTER 1] ELECTRIC CHARGE & COULOMB'S LAW**17**

P.12. A neutron is thought to be composed of one up quark of charge $\frac{2e}{3}$ and two

down quarks of charge $-\frac{e}{3}$. If down quarks are $2.6 \times 10^{-15} \text{ m}$ apart inside neutron. What is repulsive force between them? **Sol.**

Sol.**DATA**

$$\text{Charge on first down quark} = q_1 = -\frac{e}{3}$$

$$\text{Charge on second down quark} = q_2 = -\frac{e}{3}$$

$$r = 2.6 \times 10^{-15} \text{ m}$$

$$F = ?$$

Charge on first
down quark $-\frac{e}{3}$

CALCULATIONS

$$F = \frac{kq_1q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(2.6 \times 10^{-15})^2} =$$

$$= 3.78 \text{ N}$$

P.13 (a) How many electron would have to be removed from a penny

P.13 (a) How many electron would have to be removed from a penny to leave it with a charge $1.15 \times 10^{-7} \text{ C}$. (b) To what fraction of the electrons in the penny does this correspond?

Sol.**DATA**

The penny is made of copper and electrical neutral contains equal. amount of +ve and —ve charge. The magnitude of these charges can be calculated

as:

$$\text{Atomic No. of copper} = Z = 29$$

$$\text{No. of atoms in penny} = N$$

$$\text{Magnitude of charge of each atom} = Ze$$

$$\text{Let charge on penny} = NZe$$

$$\text{Standard mass of pennym} = 3.11\text{g}$$

$$\text{Molecular mass of penny} = M = 63.5 \text{ g/mole}$$

CALCULATIONS

$$\text{One mole of copper has atoms} = N_A$$

$$n \text{ moles of copper have atoms} = n N_A$$

Also

$$n \text{ moles of copper atoms} = N$$

Comparing

$$N = n N_A$$

$$N = \frac{m}{M} N_A$$

18

$$\begin{aligned} & \frac{3.11}{63.5} \times 6.02 \times 10^{23} \\ & = 2.95 \times 10^{22} \text{ atoms} \\ \text{Charge on penny} & q = 2.95 \times 10^{22} \times 29 \times 1.6 \times 10^{19} \\ & = 1.37 \times 10^5 \text{ C} \\ & q = \frac{1.37 \times 10^5}{1.6 \times 10^{19}} = 8.56 \times 10^{23} \text{ electron} \end{aligned}$$

The total charge on penny is $q = 8.56 \times 10^{23}$ electron

Charge have to retain on penny $q' = 1.15 \times 10^{-7} \text{ C}$

$$q' = \frac{1.15 \times 10^{-7}}{1.6 \times 10^{-19}} = 7.18 \times 10^{11} \text{ electron}$$

No. of electrons needed to remove from penny

$$\begin{aligned} & (8.56 \times 10^{23} - 7.18 \times 10^{11}) \text{ electrons} \\ & = 8.56 \times 10^{23} \text{ electrons} \\ & = 8.56 \times 10^{23} \times 1.6 \times 10^{19} \text{ C} \\ & = 1.3696 \times 10^{43} \text{ C} \end{aligned}$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

(b) ' Magnitude of +ve charge magnitude of —ve charge
 $1.37 \times 10^5 \text{ C}$

1.6 C charge on penny makes electrons = 1 electron
 1

IC charge on penny makes electrons = 1.6×10^{-19}

$1.37 \times 10^5 \text{ C}$ charge on penny makes electrons — $\frac{1.37 \times 10^5}{1.6 \times 10^{-19}}$ 8.56×10^{23}

Fraction of electrons = $\frac{8.56 \times 10^{23}}{7.18 \times 10^{21}}$ 1.19

P.14. An electron is in a vacuum near the surface of earth where should a second electron be placed so that the net force on first electron due to gravity is zero?

Sol.

DATA

$$q_1 = 1.6 \times 10^{-19} \text{ C}$$

$$q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$F_e = ?$$

$$r = ?$$

electron be placed so that the net force on first electron



CALCULATIONS

$$F_c = \frac{kq_1q_2}{r^2}$$

$$\frac{ke^2}{r^2} = 0 \quad \text{----- (1)}$$

$$F_g = mg$$

$$mg = 0 \quad \text{----- (2)}$$

From (1) and (2)

$$mg = \frac{ke^2}{r^2}$$

$$R = \sqrt{\frac{ke^2}{mg}} = \sqrt{\frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{9.1 \times 10^{-31} \times 9.8}}$$

$$r = 5 \text{ m}$$

P.15 Proton in cosmic rays strike the earth's atmosphere at average rate over the earth's surface, of 1500 proton/m²-s. What total current does the earth receive from beyond its atmosphere in the form of incident cosmic ray proton?

Sol.

DATA

$$A = 4\pi(R_e)^2$$

$$\frac{q}{At} = 1500 \times 1.6 \times 10^{-19} \text{ C/m}^2\text{-s} \quad [R_e = 6400 \text{ km}]$$

$$I = ?$$

CALCULATIONS

$$\frac{q}{At} = 1500 \times 1.6 \times 10^{-19}$$

$$\frac{q}{t} = 1500 \times 1.6 \times 10^{-19} \times 4\pi R_e^2$$

$$I = 0.123 \text{ A}$$

P.16. Calculate the number of coulombs of positive charge in a glass of water. Assume that volume of water to be 250 cm³.

Sol.

DATA

$$V = 250 \times 10^{-6} \text{ m}^3$$

$$\rho = 10^3 \text{ kg/m}^3$$

$$q = ?$$

CALCULATIONS

$$m = p \times v = 1000 \times 250$$

$$m = 0.25 \text{ kg} - 250 \text{ g}$$

$$18 \text{ grams of water has atoms} = 6.02 \times 10^{23} \text{ atoms}$$

$$1 \text{ gram of water has atoms} = \frac{6.02 \times 10^{23}}{18}$$

$$250 \text{g of water has atoms} = \frac{250 \times 6.02 \times 10^{23}}{18} = 8.36 \times 10^{24} \text{ atoms}$$

$$1 \text{ atom of water has +ve charge} = 10 \times 1.6 \times 10^{-19} \text{ C}$$

$$8.36 \times 10^{24} \text{ water has +ve charge} = \frac{10 \times 1.6 \times 10^{-19} \times 8.36 \times 10^{24}}{1} = 1.34 \times 10^6 \text{ C}$$

- p. 17• What equal amounts of positive charge would have to be placed on earth and on the moon to neutralize their gravitational attract. Do you need to know the moon's distance to solve this problem why and may not (a) how many metric 'tons of hydrogen would be needed to provide the positive charge calculated in part (b) the molar mass of hydrogen is 1.008 g/mol.

sol.

DATA

Charge on earth

Charge on moon

$$6 \times 10^{24} \cdot \text{kg}$$

Me

$$Mm - 7.36$$

CALCULATIONS

$$\frac{GmMe}{r^2}$$

$$6.67 \times 10^{-11}$$

$$2 \times 7.36 \times 10^{22} \times 6.0 \times 10^{24} q q 5.72 \times 10^{13}$$

(b) $m = 1.008 \text{ gTmole}$

$1.6 \times 10^{-19} \text{C}$ charge has atoms of hydrogen = 1

1

[CHAPTER 1] ELECTRIC CHARGE & COULOMB'S LAW

21

IC

$5.72 \times 10^{13} \text{C}$ charge has atoms of hydrogen = $\frac{5.72 \times 10^{13}}{1.6 \times 10^{-19}} = 3.57 \times 10^{32}$

charge has atoms of hydrogen e-T5

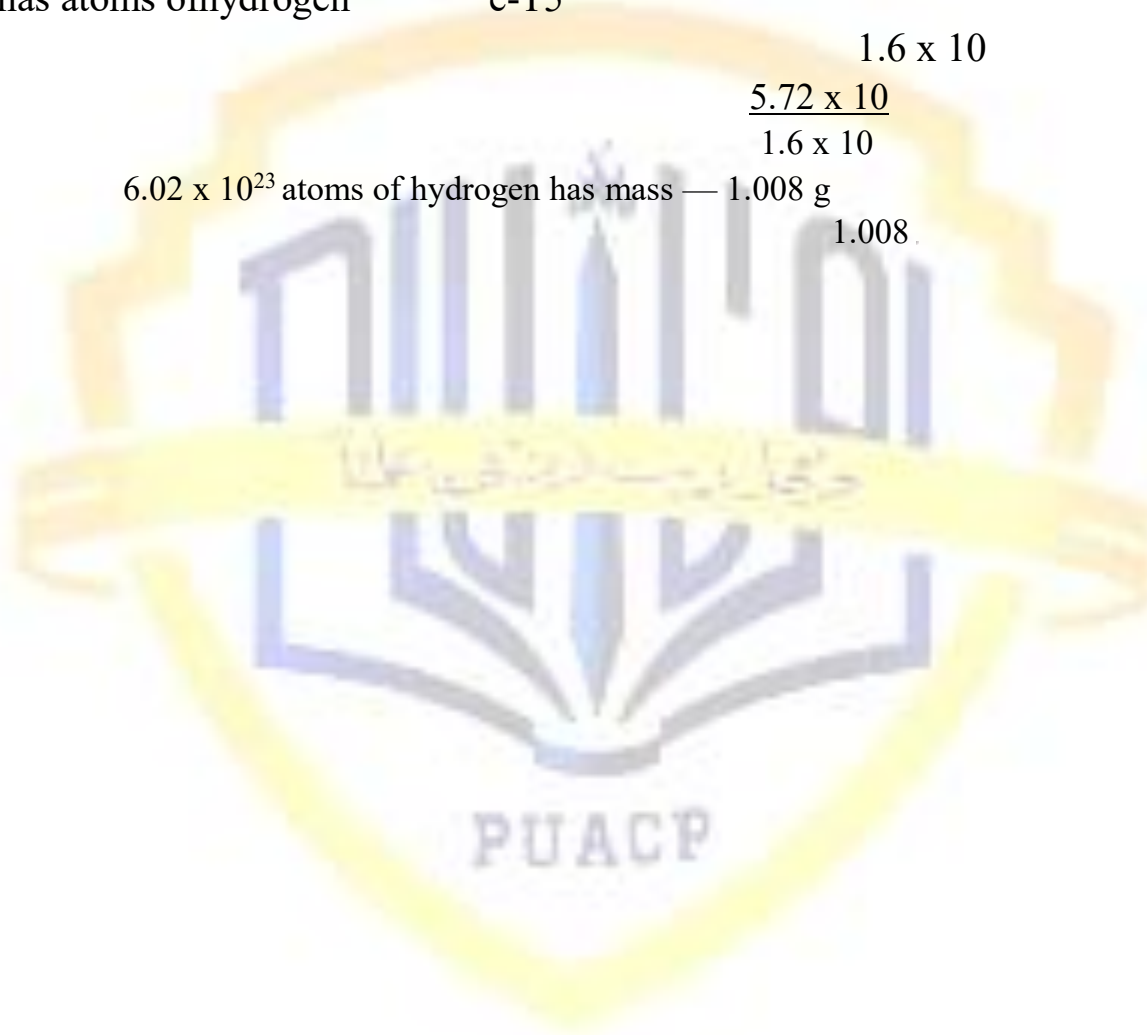
1.6×10

$\frac{5.72 \times 10}{1.6 \times 10}$

1.6×10

6.02×10^{23} atoms of hydrogen has mass — 1.008 g

1.008

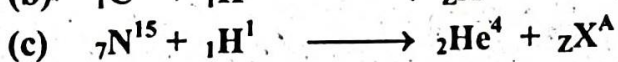
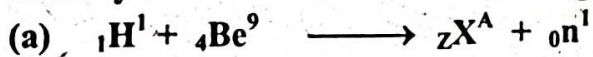


1 atom of hydrogen has mass — 6.02×10^{23}

$$\begin{aligned}
 3.57 \times 10^{32} \text{ atoms of hydrogen has mass} &= 3.57 \times 10^{32} \times \frac{1.008}{6.02 \times 10^{23}} \\
 &= 5.98 \times 10^8 \text{ g} \\
 &= 598 \text{ metric tons}
 \end{aligned}$$

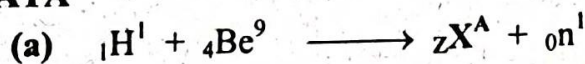
[1000 kg = metric ton]

P.18 Identify the element X in the following reaction:



Sol.

DATA



$${}_Z\text{X}^A = ?$$

CALCULATIONS

$$1 + 9 = A + 1$$

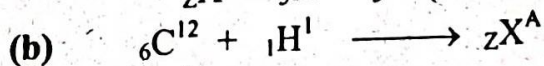
$$A = 9$$

For charge conservation

$$1 + 4 = Z + 0$$

$$Z = 5$$

$${}_Z\text{X}^A = {}_5\text{X}^9 = {}_5\text{B}^9 \text{ (element is boron)}$$



$${}_Z\text{X}^A = ?$$

CALCULATIONS

$$12 + 1 = A$$

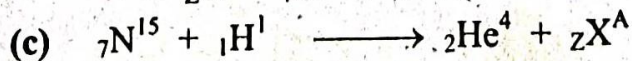
$$A = 13$$

For charge conservation

$$6 + 1 = Z$$

$$Z = 7$$

$${}_Z\text{X}^A = {}_7\text{X}^{13} = {}_7\text{N}^{13} \text{ (element is nitrogen)}$$



$${}_Z\text{X}^A = ?$$

CALCULATIONS

$$15 + 1 = 4 + A$$

$$A = 12$$

For charge conservation

$$7 + 1 = 2 + Z$$

$$Z = 6$$

$${}_Z X^A = {}_6 X^{12} = {}_6 C^{12} \text{ (element is carbon)}$$

P.19 In radioactive decay of ${}_{92}U^{238}$ the center of emerging ${}_2He^4$ particle is at certain instant 12×10^{-15} m from the center of residual ${}_{90}Th^{234}$ nucleus. At this instant (a) what is the force on the ${}_2He^4$ (b) What is acceleration of ${}_2He^4$?

Sol.

DATA

$$q_1 = 2e = 2 \times 1.6 \times 10^{-19} \text{ C}$$

$$q_2 = 90e = 90 \times 1.6 \times 10^{-19} \text{ C}$$

$$r = 12 \times 10^{-15} \text{ m}$$

$$m = 4.002 \times 1.67 \times 10^{-27} \text{ kg}$$

CALCULATIONS

(a)

$$F = \frac{kq_1q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times 2 \times 90 \times (1.6 \times 10^{-19})^2}{(12 \times 10^{-15})^2} = 288 \text{ N}$$

(b)

$$F = ma$$

$$a = \frac{F}{m}$$

$$a = \frac{288}{4.002 \times 1.67 \times 10^{-27}}$$

$$a = 4.3 \times 10^{28} \text{ m/s}^2$$



CHAPTER

R ELECTRIC FIELD

Q. 1. Define and explain electric field in detail?

ELECTRIC FIELD

The space or region around a charge in which attraction or repulsion with other charges takes place is called electric field.

The charge produces electric field which is denoted by dots. The density of dots is proportional to the strength of electric field. The presence of field is tested by test charge. The test charge is generally $+q_0$.

Consider a charge q which produces an electric field. The electric field of charge q interacts with test charge q_0 and gives rise to electric force. The electric force per unit test charge is called electric field intensity.

$$\frac{F}{q_0}$$

DIRECTION OF ELECTRIC FIELD LINES

The electric field is a vector quantity and electric field lines give its direction. The electric field lines come out from positive charge and enter into negative charge. The two electric field lines never cross each other.

UNIT

The SI unit of electric field is called N/C or V/m.

Q. 2 What are electric field lines of force? How are these lines drawn? Write their properties?

ELECTRIC FIELD LINES

The electric field lines are a map which provides information about the direction and strength of electric field at various places. The concept of electric field lines was introduced by Faraday.

Consider a $+q$ charge is placed at the centre of a sphere. Now $+q_0$ test charge placed at any point on the surface of sphere is repelled outward in radius direction. Hence, electric field lines come out from positive.

Now place $-q$ charge at the centre of sphere. The test $+q_0$ is placed at any point on the surface of sphere is attracted towards negative charge. Hence, electric field lines are radial and will enter into negative charge as shown in fig.

The map of electric field lines also provides information about the strength of electric field. The field is stronger near the charge where field lines are closer to each other.

The field lines away from charge spread out and indicating a continuous decrease in the field strength.

The number of lines per unit area passing perpendicularly through an area is proportional to the magnitude of the electric field.

The electric field lines associated with two identical positive charges are curved and seem to repel each other. The middle region shows zero field location called neutral zone,

The behavior of electric field lines of two identical negative charges is also exactly same. The fieldlines are curved which enter into negative charge. The middle region shows zero field location called neutral zone.

The electric field pattern of two opposite charges of same magnitude is shown fig. The field lines start from positive charge and end on a negative charge.

2]

The directions of resultant intensities are given by the tangents drawn to the field " lines at particular points.

The electric field lines between two oppositely charged plates are shown in fig. The middle region where field lines are parallel and equally spaced and same number of *lines passing per unit area is called uniform field.

PROPERTIES OF ELECTRIC FIELD LINES

The properties of electric field lines are as under

O The electric field lines originate from positive charges and terminate: on negative charges.

The tangent to an electric field line at any point gives the direction of the electric field at that point.

O The electric lines are closer where the electric field is strong.

The electric lines are farther apart where the electric field, is weak.

O Two electric lines of force never cross each other because E has only one direction at any given point. If the field lines cross E could have more, than one direction.

Q.3 Calculate electric field due to point charge and due to n-point charges?
--

ELECTRIC FIELD DUE TO POINT CHARGE

Consider a $+q$ charge placed at point 'O' called field charge. Place a test charge $+q_0$ at point P having distance r from charge q . The electric force between two charges is

$\vec{F}$ is called electric field intensity and denoted by $\vec{E}$.

$$\vec{E} = \frac{\vec{F}}{q}$$

The magnitude of electric field is $E = \frac{F}{q}$. Its direction is given by electric lines of force which come out from positive charge and enter, into negative charge.

ELECTRIC FIELD DUE TO IN-POINT CHARGES

Consider charges $q_1, q_2, q_3, \dots, q_n$ having distances $r_1, r_2, r_3, \dots, r_n$ from point P. The magnitude of electric field at P due to charge q_1 is

The magnitude of electric field at P due to charge q_2 is

$$E_2 = \frac{k q_2}{r_2^2}$$

and so on

The net magnitude of electric field at P due to all charges

$$E = E_1 + E_2 + E_3 + \dots + E_n$$

$$E = \frac{k q_1}{r_1^2} + \frac{k q_2}{r_2^2} + \frac{k q_3}{r_3^2} + \dots + \frac{k q_n}{r_n^2}$$

$$E = k \sum_{i=1}^n \frac{q_i}{r_i^2}$$

Q.4 Calculate electric field due to continuous charge distribution.

ELECTRIC FIELD DUE TO CONTINUOUS CHARGE DISTRIBUTION

The electric charge is quantized. The collection of large number of charges is called continuous charge distribution. The continuous charge distribution has three types.

- O Linear charge distribution
- O Surface charge distribution
- O Volume charge distribution.

ELECTRIC FIELD DUE To LINEAR CHARGE DISTRIBUTION

Consider a linear charge distribution. Take a small length element dz of the distribution having charge dq , dz length element has charge dq

Unit length element has charge λ

The charge per unit length is called linear charge density and denoted by λ

The electric field at P due to charge dq is $\frac{kX}{dz}$

The net electric field at P due to whole linear dz charge distribution is

ELECTRIC FIELD DUE TO SURFACE CHARGE DISTRIBUTION

Consider a surface charge distribution. I Take a small area element da of this distribution having charge $-dq$, da area element has charge dq

Unit area element has charge da

The charge unit area is called surface charge density is and denoted by σ

da

The electric field at P due to charge dq is $\frac{k\sigma da}{r^2}$

r

The net electric field at P due to whole surface charge distribution is $\frac{k\sigma da}{r^2}$

dq

ELECTRIC FIELD DUE TO VOLUME CHARGE DISTRIBUTION

Consider a volume charge distribution. Take a small volume element dV of this distribution having charge dq .

dV volume element has charge dq

Unit volume element has charge $= \rho dV$

The charge per unit volume is called volume charge density and denoted by ρ

ρdV

The electric field at P due to charge dq is

[CHAPTER ELECTRIC FIELD

The net electric field at P due to 'whole volume charge distribution is $E = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \hat{r}$



Q.5 What is an electric dipole? Calculate electric field at a point due to electric dipole?

ELECTRIC DIPOLE

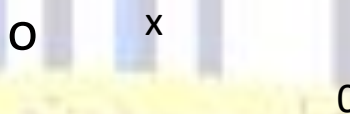
A pair of positive charge and negative charge of same magnitude having constant distance between them is called electric dipole. It is a useful concept in atoms and molecules where the effects of charge separation are measurable. It is also a useful concept in dielectrics and other applications in solid and liquid materials.

ELECTRIC DIPOLE MOMENT

The product of magnitude of charge and separation between them is called electric dipole moment and written as $P = qd$.

The electric dipole moment is a vector quantity. The direction of electric dipole moment vector is from negative charge to positive charge. **ELECTRIC FIELD DUE TO ELECTRIC DIPOLE** Consider a $+q$ and $-q$ charge having distance d between them. This is called an electric dipole. The dipole moment is given as $P = qd$

Take a point P having distance 'x' which is perpendicular bisector of distance d as shown in fig.



The magnitude of electric field at P due to $+q$ charge having distance r is

(1)

The direction of E_+ is from $+q$ charge to towards P. The magnitude of electric field at P due to $-q$ charge having distance r is

(2)

The direction of E_- is from P to towards $-q$ charge. Comparing eq(1) and eq(2) we can write as

Now to calculate net electric field at P due to dipole, resolve the electric field E_+ and E_- into components.

Rectangular components of E_+ are

$$E_{+x} = E_+ \cos \theta \quad E_{+y} = E_+ \sin \theta$$

Rectangular components of E_- are

$$E_x = E \cos \theta \quad E \quad E \sin \theta$$



[CHAPTER 2] ELECTRIC FIELD

The resultant x -component of electric field is

$$E_x = (+E_{+x}) + (-E_{-x})$$

$$E_x = E_+ \cos\theta - E_- \cos\theta$$

$$E_x = E_+ \cos\theta - E_+ \cos\theta$$

$$E_x = 0$$

The resultant y -component of electric field is

$$E_y = (-E_{+y}) + (-E_{-y})$$

$$E_y = -E_+ \sin\theta - E_- \sin\theta$$

$$E_y = -E_+ \sin\theta - E_+ \sin\theta$$

$$E_y = -2 E_+ \sin\theta$$

Magnitude of resultant electric field E is

$$E^2 = E_x^2 + E_y^2 = 0 + (-2 E_+ \sin\theta)^2$$

$$E = 2 E_+ \sin\theta$$

$$E = 2 \frac{kq}{r^2} \sin\theta$$

$$E = 2 \frac{kq}{r^2} \frac{(d/2)}{r} \quad \left(\text{From fig. } \sin\theta = \frac{d/2}{r} \right)$$

$$E = \frac{k(qd)}{r^3} = \frac{kP}{r^3}$$

$$E = kP r^{-3}$$

$$E = kP \left(x^2 + \frac{d^2}{4} \right)^{-3/2} \quad \left(\text{From fig. } r^2 = x^2 + \frac{d^2}{4} \right)$$

$$E = kP x^{-3} \left(1 + \frac{d^2}{4x^2} \right)^{-3/2} = \frac{kP}{x^3} \left(1 + \frac{d^2}{4x^2} \right)^{-3/2}$$

Expand the term in bracket by binomial theorem

$$E = \frac{kP}{x^3} \left[1 - \left(\frac{3}{2} \right) \frac{d^2}{4x^2} + \dots \right]$$

Neglect the higher order terms, we get

$$E = \frac{kP}{x^3}$$

This is the electric field at P due to electric dipole. Where x is distance between point p and bisecting point O of distance between +q and -q.

30

Q.6 Calculate the electric field due to ring of charge?

ELECTRIC FIELD DUE TO RING OF CHARGE

Consider a positively charged ring having radius R on which positive charge q is distributed uniformly. This is called linear charge distribution. Take a small length element ds of ring having charge dq .

The linear charge density λ is defined as

Now consider two length elements ds at opposite ends of a diameter of ring. We have to calculate electric field at P which is on perpendicular axis at distance z from plane of ring. From fig.

The magnitude of electric field at P due to charge element L is

$$dEL = \frac{k dq}{r^2}$$

The magnitude of electric field at P due to charge element M is

$$dEM = \frac{k dq}{r^2} \quad (2)$$

Comparing eq(1) and eq(2) we get $dEL = dEM$

Now to Calculate electric field dE at P due to both length elements, resolve the electric field dEL and dEM into components.

Rectangular components of dEL are $dELx = dEL \cos\theta$, $dELy = dEL \sin\theta$

Rectangular components of dEM are $dEMx = dEM \cos\theta$, $dEMy = dEM \sin\theta$

Resultant x —component of electric field is dEx

$$dEx = dEL \cos\theta - dEM \cos\theta = dEL \cos\theta - dEL \cos\theta = 0$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

Resultant y —component of electric field (+ dELy) is + (+ dEMy)

$$dE_y \sin\theta + dE \sin\theta \quad dE_y - dE \sin\theta + dE \sin\theta \quad dE_y - 2 dE \sin\theta$$

The magnitude of electric field dE is dE^2

$$dE^2 = dE_x^2 + dE_y^2$$

y

$$dE^2 = (0)^2 + (2 \sin\theta)^2 \quad dE = 2 \sin\theta$$

The net electric field at P due to whole ring of charge is $E = \int dE \sin\theta$

From fig. $\sin\theta = \frac{R}{r}$,

($ds = R d\theta$) Length of circumference of half ring because length elements are taken at both sides of diameter.

As $dq = \sigma ds$ so $q = \int \sigma ds = \sigma \int ds = \sigma (2\pi R)$

This is electric field at point P due to ring of charge. When point P lies at large distance $z \gg R$. The term R can be neglected as compared to z

It means ring behaves as a point charge which is concentrated at center of ring when $z \gg R$.

Q.7 Calculate electric field due to disk of charge.

ELECTRIC FIELD DUE TO DISK

Consider a positively charged circular disk of radius R having surface charge density σ . Divide the disk into small rings. Now consider such a small ring having radius r and width dr.

(esc.

Take a pair of small area element da at opposite ends of diameter denoted by L M. The area of the element having width do and length $S = (DdCL$ is given as $da = oda do$ as $da = (D du do$

The area element da has charge dq . The charge per unit area is called charge density

$$dq = \sigma da \quad du do)$$

We have to calculate electric field at point P having-distance z from the plane of the disk along its axis as shown in fig.

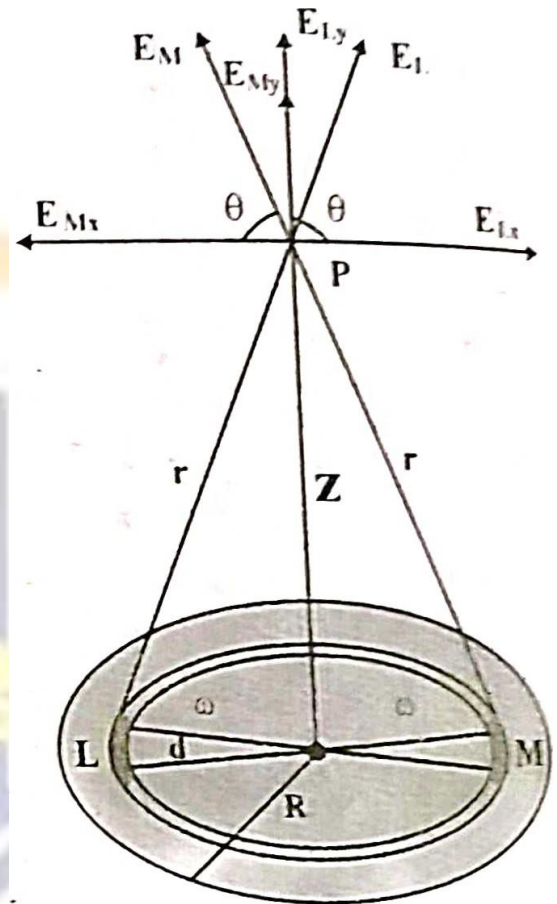
The electric field at P due to area element L having distance r is

$$dEL = \dots\dots\dots (1)$$

Similarly electric field at P due to area element M having distance r is

$$dEM = \dots\dots\dots$$

Comparing eq(1) and eq(2), we get $dEL =$



Now to calculate electric field dE at P due to both area elements, resolve electric field dEL and dEM into components.

Rectangular components of dEL are

$$dEL_x = dEL \cos\theta \quad \text{and} \quad dEL_y = dEL \sin\theta$$

Rectangular components of dEM are

$$dEM_x = dEM \cos\theta \quad \text{and} \quad dEM_y = dEM \sin\theta$$

Resultant x —component of electric field dE_x is dE_x

$$= (+ dEL_x) + (- dEM_x) =$$

$$dEL \cos\theta - dEM \cos\theta$$

$$dEL \cos\theta - dEL \cos\theta$$

$$dE_y = 0$$

2)

Resultant y —component of electric field dE_y is dE_y (+
 dEL_y) + (+ dEM_y)

33

$$dE_y = dEL \sin\theta + dEM \sin\theta$$

$$dE_y = dEL \sin\theta + dEL \sin\theta = 2 dEL \sin\theta$$

$$dEL \sin\theta$$

The magnitude of electric field dE is $dE^2 = dE_x^2 + dE_y^2 = (0)^2 + (2 dEL \sin\theta)^2$

$$dE = 2 dEL \sin\theta$$

$$dE = 2 \frac{k dq}{r^2} \left(\frac{z}{r} \right)$$

From fig. $\sin\theta =$

$$dE = 2 \frac{k z dq}{r^3} \quad \text{--- 2}$$

Use $dq = \sigma dA = \sigma 2\pi r dr$ and from fig. $r^2 = z^2 + \omega^2$

$$E = 2 \frac{k \sigma z}{(z^2 + \omega^2)^{3/2}} \int_0^R \omega d\omega$$

To calculate the resultant field due to whole disk, integrate the above eq.

$$E = \frac{k 2 \sigma z \int_0^R \omega d\omega}{(z^2 + \omega^2)^{3/2}}$$

The limit for radius of disk is from 0 to R. This R must be rotated through 2π for complete contribution of charge of disk. The limit for $d\omega$ is taken from 0 to ∞ i.e. for half circle because contribution of other half circle is already added as area element is taken at both ends of diameter.

$$E = k z (2\pi) (z^2 + \omega^2)^{-3/2} (2\pi d\omega)$$

$$E = k \sigma z (\pi) \frac{(z^2 + R^2)^{-\frac{1}{2}+1}}{2}$$

$$E = k z (\pi) \left[\frac{1}{\sqrt{z^2 + R^2}} \right]_0^R$$

(B.Sc. Classes)

$$E = -2 k \sigma z (\pi) \left[\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{\sqrt{z^2 + 0}} \right]$$

$$E = 2 \frac{1}{4\pi\epsilon_0} \sigma z (\pi) \left[\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right]$$

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right]$$

It is the electric field for $z > 0$. When $R \gg z$, the term $\frac{z}{\sqrt{z^2 + R^2}}$ goes to zero.

Then the value of electric field becomes.

280

This is electric field set up by uniform sheet of charge of infinite extent.

Q.8 Define electric field. Calculate an expression to find its intensity due to an infinite line of charge?

ELECTRIC FIELD

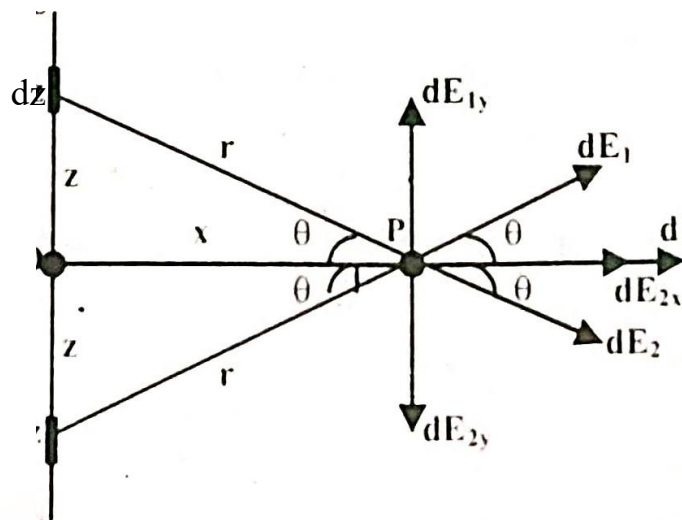
The electric force per unit charge is called electric field. The direction of the field is taken to be the direction of the force it would exert on a positive test charge. The electric field is radially outward from a positive charge and radially inward toward a negative point charge.

ELECTRIC FIELD DUE TO INFINITE LINE OF CHARGE

Consider infinite line of positive charge having linear charge density λ . Take small length element dz having charge dq . The linear charge density is

dz

Now consider a pair of small length element each of length dz on the line. Take a point P having distance x from the line joining the two length elements. We have to calculate electric field at P from the infinite charged line.



The electric field at P due to length element of bottom side having distance r is

$$dE_1 =$$

$$dE_1 = \frac{k \lambda dz}{x^2 + z^2} \quad \text{--- (1)}$$

Similarly electric field at P due to top length element having distance r is

$$dE_2 = \frac{k \lambda dz}{x^2 + z^2} \quad \text{--- (2)}$$

Comparing eq(1) and eq(2) $dE_1 = dE_2$

Now to calculate electric field dE at P due to both length elements, resolve the electric field dE₁ and dE₂ into components,

Rectangular components of dE₁ are

$$dE_{1x} = dE_1 \cos\theta \quad \text{and} \quad dE_{1y} = dE_1 \sin\theta$$

Rectangular components of dE₂ are

$$dE_{2x} = dE_2 \cos\theta \quad \text{and} \quad dE_{2y} = dE_2 \sin\theta$$

Resultant x — component of electric field dE_x is

$$dE_x = (+dE_{1x}) + (+dE_{2x})$$

$$dE_x = dE_1 \cos\theta + dE_2 \cos\theta$$

$$dE_x = dE_1 \cos\theta + dE_1 \cos\theta \quad \text{As } dE_1 = dE_2$$

$$dE_x = 2 dE_1 \cos\theta$$

Resultant y — component of electric field dE_y is

$$dE_y = (+dE_{1y}) + (+dE_{2y})$$

$$dE_y = dE_1 \sin\theta - dE_2 \sin\theta$$

$$dE_y = 0$$

$$dE_y = dE_1 \sin\theta - dE_1 \sin\theta \quad \text{[As } dE_1 = dE_2]$$

$$dE_y = 0$$

The magnitude of electric field dE is $dE = \sqrt{dE_x^2 + dE_y^2} = \sqrt{(2 dE_1 \cos\theta)^2 + (0)^2}$

$$dE = 2 dE_1 \cos\theta$$

The total electric field due to whole length is

$$E = \int_0^L 2 dE_1 \cos\theta$$

$$r k \lambda dz$$

$$E = \frac{2 \lambda}{2 \pi \epsilon_0 x} \cos \theta$$

To integrate above eq

$$\text{Put } z = x \tan \theta$$

$$dz = x \sec^2 \theta d\theta$$

$$\text{When } z = 0, \theta = 0$$

$$\text{When } z = \infty, \theta = 90^\circ = \pi/2$$

$$E = 2 k \lambda \int_0^{90} \frac{x \sec^2 \theta \cos \theta d\theta}{(x^2 + x^2 \tan^2 \theta)} = 2 k \lambda \int_0^{90} \frac{x \sec^2 \theta \cos \theta d\theta}{x^2 (1 + \tan^2 \theta)}$$

$$E = 2 k \lambda \int_0^{90} \frac{x \sec^2 \theta \cos \theta d\theta}{x^2 \sec^2 \theta} = 2 k \lambda \int_0^{90} \frac{\cos \theta d\theta}{x}$$

$$E = \frac{2 \lambda}{4 \pi \epsilon_0 x} \left[\sin \theta \right]_0^{90} = \frac{\lambda}{2 \pi \epsilon_0 x} (\sin 90^\circ - \sin 0^\circ)$$

$$E = \frac{\lambda}{2 \pi \epsilon_0 x}$$

Q.9 Define an electric dipole moment & derive an expression for torque acting on it when placed in a uniform electric field, work done by the external electric field in turning the dipole from initial angle θ_0 to final angle θ & potential energy.

ELECTRIC DIPOLE

A pair of positive charge and negative charge of same magnitude having constant distance between them is called dipole.

ELECTRIC DIPOLE MOMENT

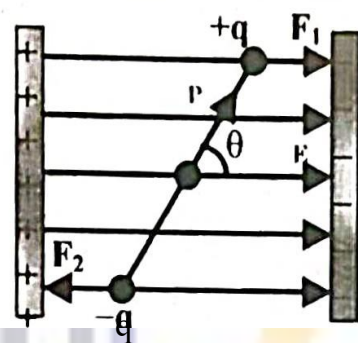
[CHAPTER ELECTRIC FIELD

The product of magnitude of charge and separation between them is called dipole moment and written as $P = qd$. It is a vector quantity. The direction of dipole moment vector is from negative charge to positive charge.

MOTION OF ELECTRIC DIPOLE IN ELECTRIC FIELD

Consider a $+q$ and $-q$ charge having distance d between them is placed in external electric field E . The electric dipole moment vector P makes angle θ with electric field E as shown in fig.

The electric force $\vec{F}_1$ acts on charge $+q$ and electric force $\vec{F}_2$ acts on charge $-q$. This couple tends to rotate the electric dipole moment vector about its center of mass in clockwise direction.
The electric torque $\vec{\tau}_e$ generated due to couple tends to align the $\vec{P}$ vector with $\vec{E}$ vector.



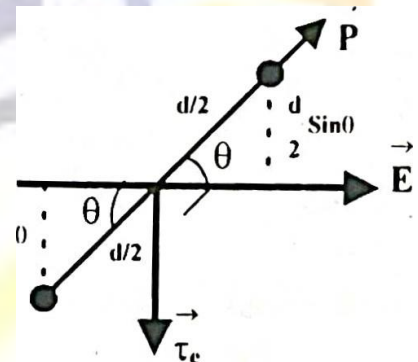
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The magnitude of net electric torque acting on dipole moment vector is

$$\tau = F \frac{d}{2} \sin\theta + F \frac{d}{2} \sin\theta$$

$$\tau = F d \sin\theta \quad \tau = q E d \sin\theta \quad [F = qE]$$

$$\tau = P E \sin\theta$$



The electric torque in vector form is

The direction of electric torque is given by right hand rule which is directed into plane of paper.

When P is aligned with E , Rotate vector P through angular displacement $d\theta$ anticlockwise to give some angular displacement. The direction of $d\theta$ is out of plane of paper according to right hand rule. Now vector P when released rotates clockwise due

to torque and again tends to align with E . Hence angle between electric torque and angular displacement is 180° .

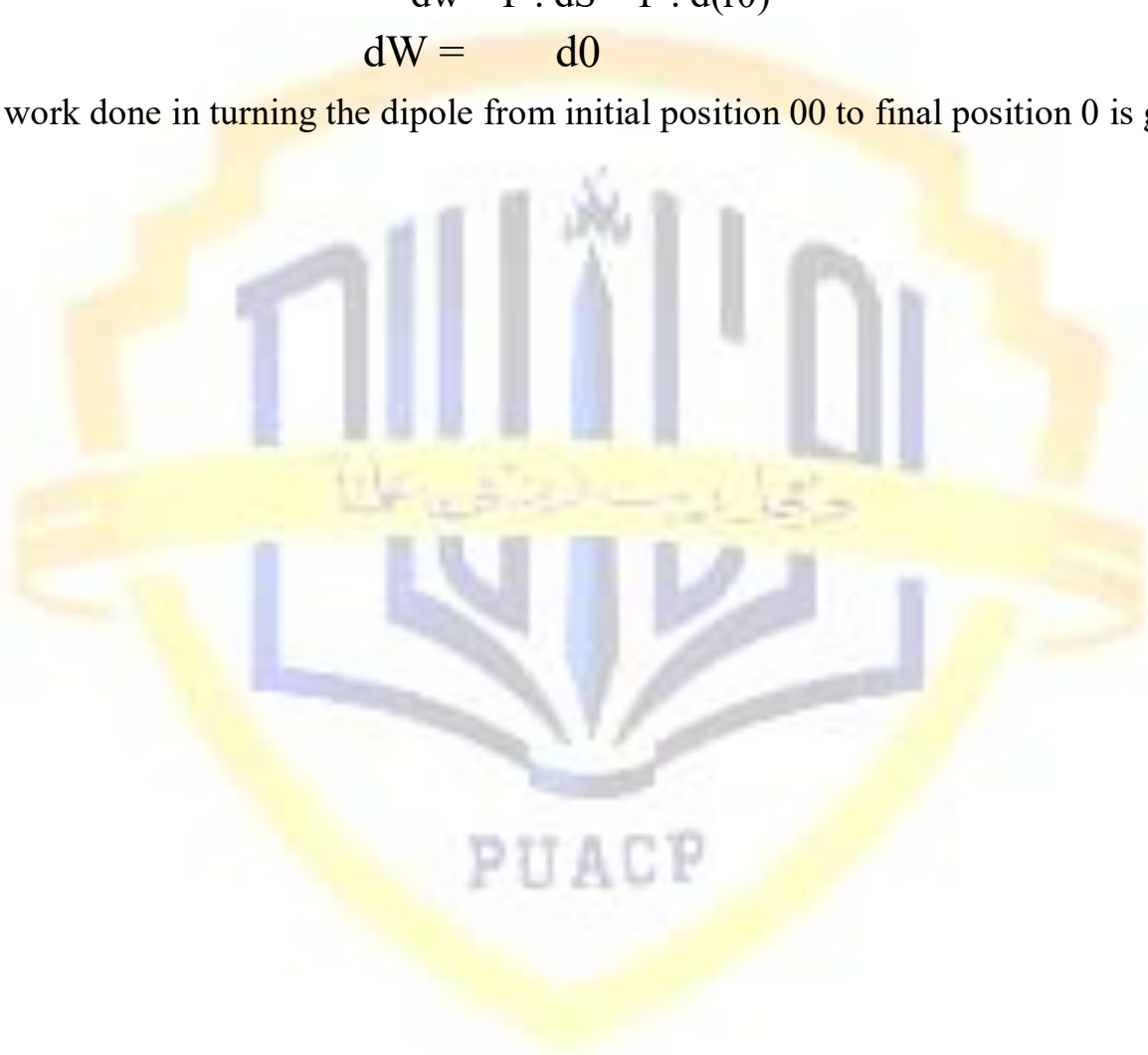
The electric field produces a torque on a dipole which tends to take it to its low energy configuration. Therefore, its rotation from the low energy state against the field requires work.

The amount of work done by electric torque in electric field in turning the electric dipole through small angle $d\theta$ is

$$dw = \tau \cdot d\theta = \tau \cdot d(\cos\theta)$$

$$dW = -\tau \sin\theta d\theta$$

∴ The net work done in turning the dipole from initial position θ_0 to final position θ is given as



$$W = \int_{0}^{\theta} \tau_e \cdot d\theta$$

$$W = \int_{0}^{\theta} p E \sin\theta \, d\theta = -p E \cos\theta \Big|_0^{\theta} = -p E (\cos\theta - \cos 0)$$

The direction of torque and angular displacement is given by right hand rule. Both are anti parallel to each other,

$$W = -p E \cos\theta$$

$$W = p E \cos\theta$$

$$W = p E (\cos\theta - \cos 0)$$

POTENTIAL ENERGY

According to work energy theorem, change in potential energy is equal to work done.

$$\Delta U = -W$$

Consider that initial position of dipole vector is at $\theta = 90^\circ$. The potential energy at this position is zero. The potential energy at any position θ is

$$U(\theta) - U(90^\circ) = -p E (\cos\theta - \cos 90^\circ)$$

$$U(\theta) - 0 = -p E (\cos\theta - 0)$$

$$U(\theta) = -p E \cos\theta$$

The potential energy will be minimum when electric field E and dipole vector P are parallel i.e. $\theta = 0^\circ$

$$U(\theta) = -\vec{P} \cdot \vec{E}$$

$$U_{\min} = -p E \cos 0^\circ = -p E$$

The potential energy will be maximum when electric field E and dipole moment vector P are anti parallel i.e. $\theta = 180^\circ$

$$U_{\max} = -p E \cos 180^\circ = +p E$$

SHORT QUESTIONS

Q:] A positively charged ball hangs from a long silk thread. We wish to measure E at a point in the same horizontal plane as that of the hanging charge. To do so we put a positive test charge (q_0) at that point and measure F/q_0 . Will F/q_0 be less than, equal to or greater than E at that point.

Ans. • The electric field E at a point is given as $E = F/q_0$. Hence F/q_0 is equal to E .

Q:2 In exploring electric fields with a test charge, we have often assumed, for convenience that the test charge was positive. Does this really make any difference in determining the field?

Ans: It makes no difference on magnitude of electric field while exploring electric fields with positive test charge but direction is given as.

If test charge is positive the direction of electric field lines is given as "electric field lines come out from positive charge and enter into negative charge.

If test charge is negative the direction of electric field lines is given as "electric field lines come out from negative charge and enter into positive charge. Q:3 Electric lines of force never cross each other why.

Ans: The concept of electric lines of force was given by Faraday. He assumed that $+q$ charge is placed at the center of a circle. When test charge $+q_0$ is placed at any point on circumference of circle, the test charge moves outward along radius direction due to repulsive force. Therefore electric field lines act along radius of circle. The two radii of a circle never cross each other so electric field lines also never cross each other.

The electric field lines are unidirectional. For curved field lines, the tangent to a field line at any point gives the direction of the electric field line at that point. No two lines cross each other. This is because E has only one direction at any point. If the lines cross, E could have more than one direction.

Q:4 A point charge is moving in an electric field at right angles to the electric lines of force. Does any force act on it.

(CHAPTER 21 ELECTRIC FIELD

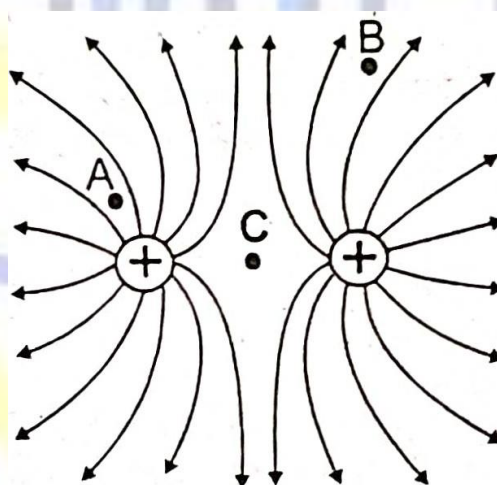
Ans: The electrostatic force acts on charge when a point charge is moving in an electric field at right angle to the electric lines of force because the electric force is given as $q E$. It does not depend upon angle.

Q:5 What is the origin of electrostatic cling "n phenomenon that some tinnce affects clothes as they are removed from a dryer?

Ans: The origin of electrostatic cling is the charges which are produced on a body due to sharing of electrons. The charges are produced due to friction.

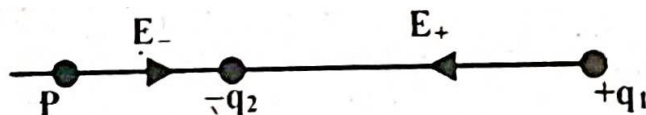
Q:6 Two-point charges of unknown magnitude and sign are at a distance d apart. The electric field is zero at one point between them on the line joining them. What can you conclude about the charges?

Ans: The electric field vanishes at one point on the line joining two point charges q_1 and q_2 which are of the same sign irrespective of their magnitudes. It reveals that lines in the region between two like charges seem to repel each other. The middle region shows the neutral zone.



Q:7 Two-point charges of unknown magnitude and sign are at a distance d apart. (a) Is possible to have $E = 0$ at any point not between the charges but on the line joining them, what are the necessary conditions and where is the point located. (b) Is it possible for any arrangement of two point charges to find two points at which $E = 0$ if so under what conditions.

Ans: Consider two point charges $+q_1$, and $-q_2$ such that $-q_2$ is smaller than $+q_1$. These charges are placed on a line and separated by distance d . The electric field lines come out from positive charge towards point P, The electric field lines enter into negative charge from point P.



The electric field will be zero at a point P which lies on left side of charge $-q_2$ when $E_+ = E_-$. This mathematical arrangement will give you a quadratic eq. The solution of this eq gives distance of point P from charge $-q_2$.

Q:8 A point charge q of mass m is released from rest in non-uniform electric field.

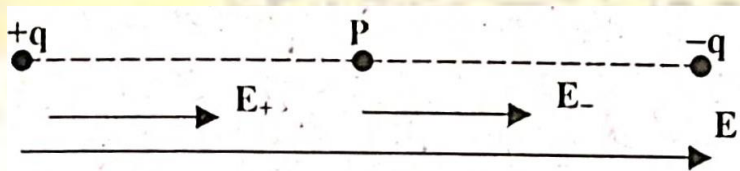
(a) Will it necessarily follow the line of force that passes through the release point? (b) Under what circumstances if any, will charged particle follow the electric field lines?

Ans: (a) When a point Charge q of mass m is released from rest in non-uniform electric field, it will not follow the line of force that passes through the release point.

(b) The charged particle can follow the electric field lines that pass through the release point if electric field is uniform.

A positive and negative charge of the same magnitude lie on a long straight line. What is the direction of E for points on this line that lie between the charges?

Ans: The direction of E for points between a positive and negative charge of the same magnitude lie on a long straight line is from positive charge towards negative charge.



Q:10 Compare the way, E varies with r for (a) a point charge (b) dipole and (c) quadrupole

Ans: The electric field due to point charge is $E = \frac{kq}{r^2}$ The electric field due to dipole

is $E = \frac{k p \cos \theta}{r^3}$ This shows that the electric field varies with r^{-2} for point charge, r^{-3} for dipole. By symmetry r^{-4} for quadrupole.

Q:11 For what orientations of an electric dipole in a uniform electric field, is the potential energy of the dipole (a) the greatest (b) the least.

Ans. The potential energy of dipole is given as

High energy

and θ are anti-parallel.

Put $\theta = 180^\circ$.

(CHAPTER 21 ELECTRIC FIELD

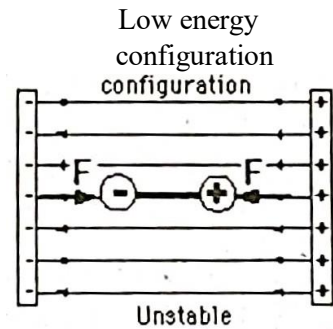
$$U = -P E \cos 180^\circ$$

$$U = -\vec{P} \cdot \vec{E} \quad \text{configuration}$$

$$U = P E \cos \theta$$

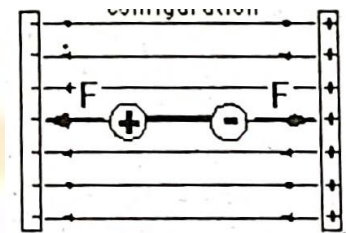
The potential energy of dipole is the greatest when $\theta = 180^\circ$

$$U = -PE$$



The potential energy of dipole is the least when $\theta = 0^\circ$ and $\vec{P}$ and $\vec{E}$ are parallel. Put $\theta = 0^\circ$.

$$U = P E \cos 0^\circ$$



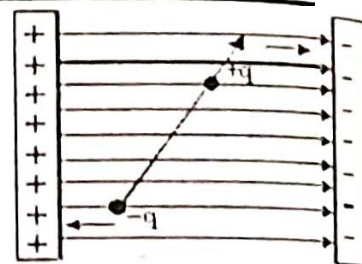
Q:12 An electric dipole is placed in non-uniform electric field. $U = -\vec{P} \cdot \vec{E}$ Is there net force act on it?

Ans: Yes, the net electric force acts on a dipole when an electric dipole is placed in non-uniform electric field.

Q:13 An electric dipole is placed at rest in a uniform external electric field shown in fig. and released. Discuss its motion.

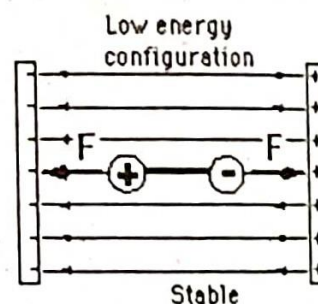
Ans: The motion of dipole will be clockwise because dipole moment $\vec{P}$ tends to align

with electric field E . The potential energy of the system tends to a minimum when the dipole moment is $+$ aligned with external field. After that dipole may start oscillations due to inertia.



Q:14 An electric dipole has its dipole moment P aligned with a uniform external electric field E (a) is the equilibrium stable or unstable (b) Discuss the nature of the equilibrium if P and E point in opposite direction.

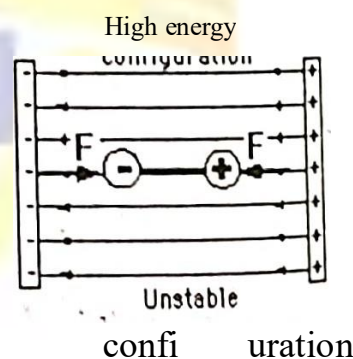
Ans: (a) The equilibrium will be Stable when an electric dipole moment P is aligned with a uniform external electric field E . This is low energy configuration.



(b) When an electric dipole moment P is aligned with uniform external electric field E in opposite direction, the dipole moment vector

P tends to rotate towards E through 180° . This is high-

the dipole moment vector



energy configuration. The equilibrium is unstable. After

that P tends to align with E to get stable equilibrium. Then it starts oscillations between unstable and stable equilibrium positions.

Q: 15 You turn an electric dipole end for end in a uniform electric field. How does the work you do depend on the initial orientation of the dipole with respect to field.

Ans: The amount of work done by external electric field in turning the dipole from an initial angle θ_0 to a final angle θ is

$$W = P E (\cos\theta - \cos\theta_0)$$

(CHAPTER 21 ELECTRIC FIELD

As an electric dipole is turned end for end in a uniform electric field, Put $\theta = 180^\circ$ and $\theta = 0^\circ$

$$W = P E (\cos\theta - \cos\theta_0)$$

$$W = P E (1 - (-1))$$

$$W = 2 P E$$

Q:16 Give four properties of electric field lines.

Ans. The four properties of electric field lines are

Electric field lines originate from positive charge and end on negative charge. The tangent to an electric field line at any point gives the direction of the electric field at that point.

The lines are closer where the electric field is strong and the lines are farther apart where the electric field is weak.

O No two lines cross each other. This is because E has only one direction at any point. If the lines cross, E could have more than one direction.

Q:17 An atom is normally electrically neutral. Why then should an a particle be deflected by the atom under any circumstances?

Ans: The positive charge of atom is densely concentrated at the center of the atom called nucleus. The electrons are revolving in orbits around nucleus. When positively charged α -particle is bombarded on an atom, it is deflected due to nucleus.

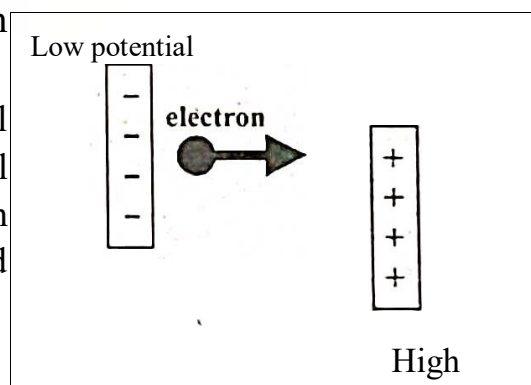
Q:18 Mention only the names of different types of charge densities used to describe continuous distribution of charge.

Ans: The names of different types of charge densities used to describe continuous distribution of charge are linear charge distribution, surface charge distribution

and volume charge distribution.

Q:19 Do electron tends to go region of high potential or low potential?

Ans: The electron tends to go from low potential region to high potential region. The low potential (O side pushes negatively charged electron and high potential (+) side attracts negatively charged electron.



Q:20 Write in your own words the purpose of Milliken oil drop experiment.

Ans. The purpose of Milliken oil drop experiment was to measure the charge on electron.

Q:21 Name three types of charge densities used to describe continuous charge distribution?

Ans. The names of three charge densities used to describe continuous charge distribution are called

• LINEAR CHARGE DENSITY

- The charge per unit length is called linear charge density



The SI unit of linear charge density is called C m

2nd SURFACE CHARGE DENSITY

The charge per unit area is called surface charge density

da

The SI unit of surface charge density is called C m⁻²

3rd VOLUME CHARGE DENSITY

The charge per unit volume is called volume charge density

$$\rho = \frac{dq}{dV}$$

The SI unit of volume charge density is called C/m³

SAMPLE PROBLEMS

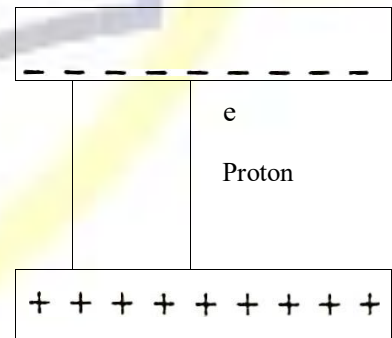
P.1. A proton is placed in uniform electric field E. What must be the magnitude and direction of electric field. If electrostatic force is balanced by weight?

Sol.

DATA

$$m_p = 1.67 \times 10^{-27} \text{ kg } q = 1.6 \times 10^{-19} \text{ C}$$

CALCULATIONS



q

$$F = \frac{1.67 \times 10^{-27} \times 9.8}{1.6 \times 10^{-19}}$$

-19

$1 \times 10^{-7} \text{ N/C}$

The electric field E is directed against gravity force

P.2 In an ionized He atom (one electron removed) the electron and nucleus are separated by 26.5 pm. What is the field due to nucleus at the location of electron?

Sol.

2) ELECTRIC FIELD**45**DATA $q = 2 \times 1.6 \times 10^{-19} \text{ C}$ $r = 26.5 \times 10^{-12} \text{ m}$

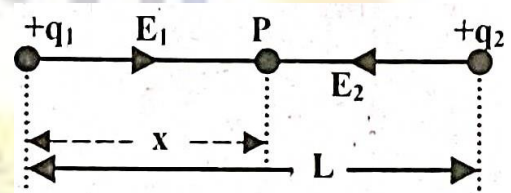
$$E = \frac{kq}{r^2}$$

$$\text{CALCULATIONS } E = \frac{9 \times 10^9 \times 2 \times 1.6 \times 10^{-19}}{(26.5 \times 10^{-12})^2} = 4.10 \times 10^{12} \text{ N/C}$$

p.3 A charge $q_1 = +1.5 \text{ } \mu\text{C}$ and a charge $q_2 = +2.3 \text{ } \mu\text{C}$. The first charge is at origin of an x-axis and second is at position $x = L$. Where $L = 13 \text{ cm}$. At what point P along x-axis is the electric field zero?

Sol.

DATA

 $q_1 = 1.5 \text{ } \mu\text{C}$ $q_2 = 2.3 \text{ } \mu\text{C}$ $L = 13 \text{ cm}$ x ? (zero field location)**CALCULATIONS**Electric field at P due to $+q_1$ charge

$$E_1 = \frac{kq_1}{x^2} = \frac{k(1.5 \text{ } \mu\text{C})}{x^2}$$

 E_1 is directed from $+q_1$ charge towards PElectric field at P due to $+q_2$ charge

$$E_2 = \frac{kq_2}{(L-x)^2} = \frac{k(2.3 \text{ } \mu\text{C})}{(L-x)^2}$$

Electric field at P due to $+q_2$ charge E_2 is directed from $+q_2$ charge towards P

For zero field location at P

$$\frac{k(1.5 \text{ } \mu\text{C})}{x^2} = \frac{k(2.3 \text{ } \mu\text{C})}{(L-x)^2}$$

2.3

$$1.5 \times 10^{-4} (13 - x)^2$$

46

$$(13 - x)^2 = 2.315 \times 10^{-2}$$

$$13 - x = 1.23 \times 10^{-1}$$

$$2.23 \times 10^{-1} = 13$$

$$13 - x = 2.23 \times 10^{-1}$$

$$5.8 \text{ cm}$$

- P.4** A molecule of water vapour has an electric dipole moment $P = 6.2 \times 10^{-30} \text{ C}\cdot\text{m}$. (a) How far apart are the effective centers of positive and negative charge in a molecule of H_2O ? (b) What is the maximum torque on a molecule of H_2O in electric field of $1.5 \times 10^4 \text{ N/C}$. (c) Suppose that dipole moment of a molecule of H_2O is initially pointing in a direction opposite to the field. How much work is done by the electric field in rotating the molecule into alignment with the field.

Sol.

DATA

$$\text{No. of electrons in H}_2\text{O} = 10 \quad q = 1.6 \times 10^{-19} \text{ C}$$

$$P = 6.2 \times 10^{-30} \text{ C}\cdot\text{m}$$

$$E = 1.5 \times 10^4 \text{ N/C}$$

$$(a) \quad d = ?$$

$$(b) \quad \tau_c =$$

$$(c) \quad W =$$

CALCULATIONS

$$6.2 \times 10^{-30} \times 10 \times 1.6 \times 10^{-19} \text{ (d)}$$

$$\tau = P E \sin \theta$$

$$\tau_c = 6.2 \times 10^{-30} (1.5 \times 10^4) \sin 90^\circ$$

$$\tau = 9.3 \times 10^{-26} \text{ N}\cdot\text{m}$$

$$(c) \quad \text{from } 180^\circ \text{ to } 0^\circ = 0.$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

$$d = 3.9 \times 10^{-12} \text{ m}$$

(b) o 900

$$w (\cos \theta - \cos \theta_0) = 6.2 \times 10^{-30} \times 1.5 \times 10^{-4} (\cos \theta - \cos 180^\circ) = 1.9 \times 10^{-33} \text{ J}.$$



EXERCISE PROBLEMS

p.] An electron is accelerated eastward at $1.84 \times 10^9 \text{ m/s}^2$ by electric field. Determine the magnitude and direction of electric field?

sol.

DATA

$$a = 1.84 \times 10^9 \text{ m/s}^2 \quad q = 1.6 \times 10^{-19} \text{ C}$$

-31

$$m = 9.1 \times 10^{-31} \text{ kg}$$

CALCULATIONS

$$\frac{9.1 \times 10^{-31} \times 1.84 \times 10^9}{1.6 \times 10^{-19}}$$

$$0.01 \text{ N/C}$$

E is along westward.

P.2 Calculate the magnitude of electric field due to electric dipole moment

$3.56 \times 10^{-29} \text{ C}\cdot\text{m}$ at a point 25.4 nm away along bisector axis?

Sol.

DATA

$$P = 3.56 \times 10^{-29} \text{ C}\cdot\text{m}$$

$$25.4 \times 10^{-9} \text{ m}$$

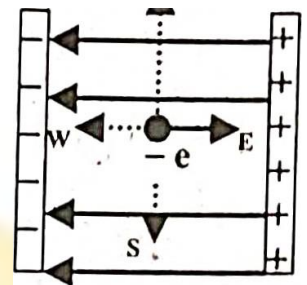
=9

CALCULATIONS.

$$E = \frac{kP}{x^3}$$

$$E = \frac{9 \times 10^9 \times 3.56 \times 10^{-29}}{(25.4 \times 10^{-9})^3} = 19552 \text{ N/C}$$

AN



P.3 An infinite- line of charge produces a field of $4.52 \times 10^4 \text{ N/C}$ at a distance of 1.96 m . Calculate the linear charge density.

sol.

DATA

$$E = 4.52 \times 10^4 \text{ N/C}$$

$$x = 1.96 \text{ m}$$

48

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

CALCULATIONS

$$E = \frac{\lambda}{2 \pi \epsilon_0 x}$$

$$\lambda = E (2 \pi \epsilon_0 x)$$

$$\lambda = (4.52 \times 10^4 \times 2 \pi \times 8.85 \times 10^{-12} \times 1.96)$$

$$\lambda = 4.14 \times 10^{-14} \text{ C/m}$$

P.4 Calculate dipole moment of electron and proton 4.3 nm apart?

Sol.

DATA

$$d = 4.3 \times 10^{-9} \text{ m}$$

$$q = +1.6 \times 10^{-19} \text{ C} = \text{Charge of proton}$$

$$q = -1.6 \times 10^{-19} \text{ C} = \text{Charge of electron}$$

q

-1.6

$$P = qd$$

CALCULATIONS

$$P = qd$$

$$P = 1.6 \times 10^{-19} \times 4.30 \times 10^{-9}$$

$$P = 6.88 \times 10^{-28} \text{ C-m}$$

P.5 Humid air breaks in electric field of $3 \times 10^6 \text{ N/C}$. What is the magnitude of the electric force on electron and for ion.

Sol.

DATA

$$E = 3.0 \times 10^6 \text{ N/C} \quad q = 1.6 \times 10^{-19} \text{ C}$$

CALCULATIONS

. For electron .

$$F = 1.6 \times 10^{-19} \times 3 \times 10^6 = 4.8 \times 10^{-13} \text{ N}$$

For ion .

$$F = qE$$

$$F = 1.6 \times 10^{-19} \times 10^6 = 1.6 \times 10^{-13} \text{ N}$$

CHAPTER 21 ELECTRIC FIELD

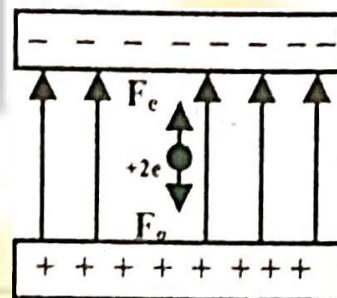
EXERCISES (I.U.D.)

- P.6 An α -particle, the nucleus of He atom, has mass of $6.64 \times 10^{-27} \text{ kg}$ and charge $+2e$. What are the magnitude and direction of electric field that will balance its weight?

sol.

$$m = 6.64 \times 10^{-27} \text{ kg}$$

$$q = 2 \times 1.6 \times 10^{-19} \text{ C}$$



CALCULATIONS

For balanced condition

$$qE = mg$$

$$E = \frac{mg}{q} = \frac{6.64 \times 10^{-27} \times 9.8}{2 \times 1.6 \times 10^{-19}} = 2.03 \times 10^{-8} \text{ N/C}$$

Electric field is directed against gravity direction.

- P.7 In uniform electric field near the surface of the Earth, a particle having charge $-2 \times 10^{-10} \text{ C}$ is acted on by a downward force $6.64 \times 10^{-27} \text{ N}$. (a) Find the magnitude of electric field. (b) What is the magnitude and direction of force exerted on proton placed in that field, (c) What is ratio of electric to gravitational force?

Sol.

$$\text{DATA (a) } q = 2.0 \times 10^{-10} \text{ C}$$

$$F = 6.64 \times 10^{-27} \text{ N}$$

CALCULATIONS

$$E = \frac{F}{q} = \frac{6.64 \times 10^{-17}}{2.0 \times 10^{-19}} = 3.32 \times 10^{18} \text{ N/C}$$

(b) For proton $q = 1.6 \times 10^{-19} \text{ C}$

$\times 10^{-17}$

$$E = 3.32 \times 10^{18} \text{ N/C}$$

The force acts along the

3.32×10^{18} $5.31 \times 10^{-37} \text{ N}$
of field

(c) Ratio of electric to gravitational force ($F_g = mg$)

$$5.31 \times 10^{-37}$$

$$3.24 \times 10^{-37}$$

$$1.67 \times 10^{-27} \times 9.8$$

P.8 What is magnitude of point charge, chosen so that electric field 75 cm away has value 2.30 N/C ?

Sol.

DATA

$$r = 75 \times 10^{-2} \text{ m}$$

$$E = 2.30 \text{ N/C}$$

CALCULATIONS

$$\begin{aligned} 2.30 &= \frac{(9 \times 10^9)q}{(75 \times 10^{-2})^2} \\ &= 1.4 \times 10^{-10} \text{ C} \end{aligned}$$

P.9 At what distance along the axis of charged disk of radius R is the field equals to one-half the field at surface of disk at centre?

Sol.

Electric field at P due to charged disk is

$$E_P = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{(z^2 + R^2)^{1/2}} \right]$$

Electric field at surface of disk for which $z = 0$

$$E = \frac{\sigma}{2\epsilon_0} [1 - 0] = \frac{\sigma}{2\epsilon_0}$$

The given condition is that

$$E_p = \frac{1}{2} E$$

$$\frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{(z^2 + R^2)^{1/2}} \right] = \left(\frac{1}{2} \right) \frac{\sigma}{2\epsilon_0}$$

$$\frac{z}{(z^2 + R^2)^{1/2}} = 1 - \frac{1}{2} =$$

$$4z^2 = z^2 + R^2$$

[CHAPTER 2] ELECTRIC FIELD

51

p.10 Two apart equal (a) what and opposite is magnitude charges of of electric magnitude field at 1.88×10^{-2} point? 'C are (b) held What 15.2 forcecm

would act on electron placed at there?

sol.

DATA

$-1.88 \times 10^{-7} \text{ C}$

$q_2 = +1.88 \times 10^{-7} \text{ C}$

$15.2 \times 10^{-2} \text{ m}$

CALCULATIONS

The electric at P(mid) due to $(-q_1)$ charge is field at P(mid)

$$E_1 = \frac{kq_1}{(d/2)^2}$$

cted towards $(-q_1)$ from point P.

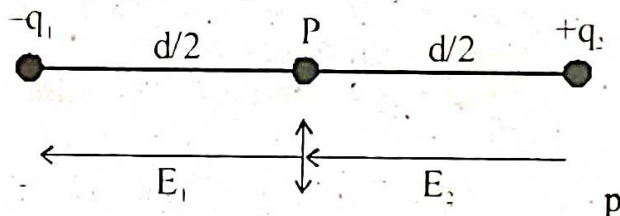
The electric field is to q_2 directed. The electric field at P due charge is E_2

$$E_2 = \frac{kq_2}{(d/2)^2}$$

$$E = E_1 + E_2$$

The net field is

$$9 \times 10^9 \times \frac{1.88 \times 10^{-7}}{(15.2 \times 10^{-2})^2} + 9 \times 10^9 \times \frac{1.88 \times 10^{-7}}{(15.2 \times 10^{-2})^2} \times 10$$



$$E = 5858.72 \text{ N/C}$$

E is directed towards +q₂ to -q₁

$$(b) \quad F = -1.6 \times 10^{-19} \times 5858.72$$

$$F = -9.37 \times 10^{-16} \text{ N}$$

P.11 In Milliken's experiment a drop of radius 1.64 μm and density $\rho = 0.851 \text{ g/cm}^3$ is balanced when electric field of $1.92 \times 10^5 \text{ N/C}$ is applied. Calculate magnitude of charge in terms of e?

Sol.

DATA: $q = 9$

$$\rho = 0.851 \text{ g/cm}^3 = 851 \text{ kg/m}^3, R = 1.64 \times 10^{-6} \text{ m}$$

$$E = 1.92 \times 10^5 \text{ N/C}$$

52

CALCULATIONS

For balanced condition

$$qE = mg$$

$$q =$$

$$= \frac{4\pi R^3 \rho g}{3E} = \frac{4\pi (1.64 \times 10^{-6})^3 \times 9.8 \times 851}{3 \times 1.92 \times 10^5}$$

$$q = 8.02 \times 10^{-19} \text{ C}$$

Electric-field is directed downward hence charge must be negative so that electric force acts upward to balance weight. $q = n(e)$

$$8.02 \times 10^{-19} = n(1.6 \times 10^{-19})$$

$$n = 5$$

P.12 An electric dipole consists of charges of magnitude 1.48 nC separated by 6.23 cm. It is in electric field of 1100 N/C (a) Calculate the magnitude of dipole moment (b) What is difference in PE corresponding to dipole orientation parallel and anti-parallel to field?

Sol.

$$\text{DATA: } d = 6.23 \times 10^{-2} \text{ m}, q = 1.48 \times 10^{-9} \text{ C}$$

$$E = 1100 \text{ N/C}$$

U = ? (Parallel and anti-parallel)
CALCULATIONS

$$p = 1.48 \times 6.23 \times 10^{-30} \text{ C m}$$

(b) For parallel condition of dipole and field

$$U = -p E \cos \theta$$

$$U = -9.2 \times 10^{-30} \times 1.1 \times 10^6 \times \cos 0^\circ = -1.012 \times 10^{-24} \text{ J}$$

For anti-parallel condition of dipole and field

$$U = -p E \cos 180^\circ$$

$$U = -9.2 \times 10^{-30} \times 1.1 \times 10^6 \times \cos 180^\circ$$

$$U = 1.012 \times 10^{-24} \text{ J}$$

[CHAPTER

21

ELECTRIC FIELD p.13 An electric dipole consists of charges $+2e$, $-2e$ separated by 0.78 nm . It is in electric field of $3.4 \times 10^6 \text{ N/C}$. Calculate the magnitude of torque on the dipole when dipole moment (a) parallel (b) right angle (c) opposite to field.

Sol.

DATA $d = 0.78 \times 10^{-9} \text{ m}$

$q =$

$$= +2e = 2 \times 1.6 \times 10^{-19} \text{ C}$$

$$-2e = -2 \times 1.6 \times 10^{-19} \text{ C}$$

$$E = 3.4 \times 10^6 \text{ N/C}$$

$\tau_e = ?$ (parallel, right angle and opposite to field)

CALCULATION

(a) When dipole moment is parallel ($\theta = 0$)

$$\tau_e = p E \sin \theta = q d E \sin 0^\circ = 0$$

(b) When dipole moment is at right angle $\theta = 90^\circ$

$$\tau_e = q d E \sin 90^\circ$$

$$\tau_e = 2(1.6 \times 10^{-19}) \times 0.78 \times 10^{-9} \times 3.4 \times 10^6$$

$$\tau_e = 8.4 \times 10^{-22} \text{ N-m}$$

(c) When dipole moment is opposite to field ($\theta = 180^\circ$)

$$\tau_e = q d E \sin 180^\circ = 0$$

P.14 Find the frequency of oscillation of an electric dipole moment P and rotational inertia I for small amplitude of oscillation about its equilibrium position in uniform electric field E . sol.

ELECTRICITY AND MAGNETISM (B.Sc.

$$\tau_e = P E \sin \theta \quad \text{----- (1)} \quad \tau_e = r F$$

$$\sin \theta \quad \text{----- (2)}$$

Comparing eq. (1) and eq. (2)

$$P E \sin \theta = r F \sin \theta \quad P E = r m a$$

$$[F = m a]$$

$$m r (r \alpha)$$

$$m r^2 \alpha = m l^2 \omega$$

$$\frac{P E}{l \omega}$$

